# Modelling Animal Movement For Behavioural Inference


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BioSS

# —Background

Some marine animals are less protected at sea from human activities than on land.

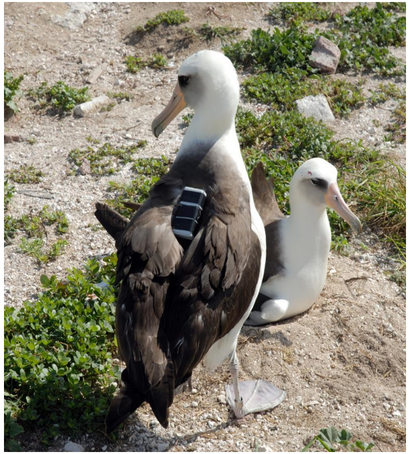
Offshore wind farms

Overfishing

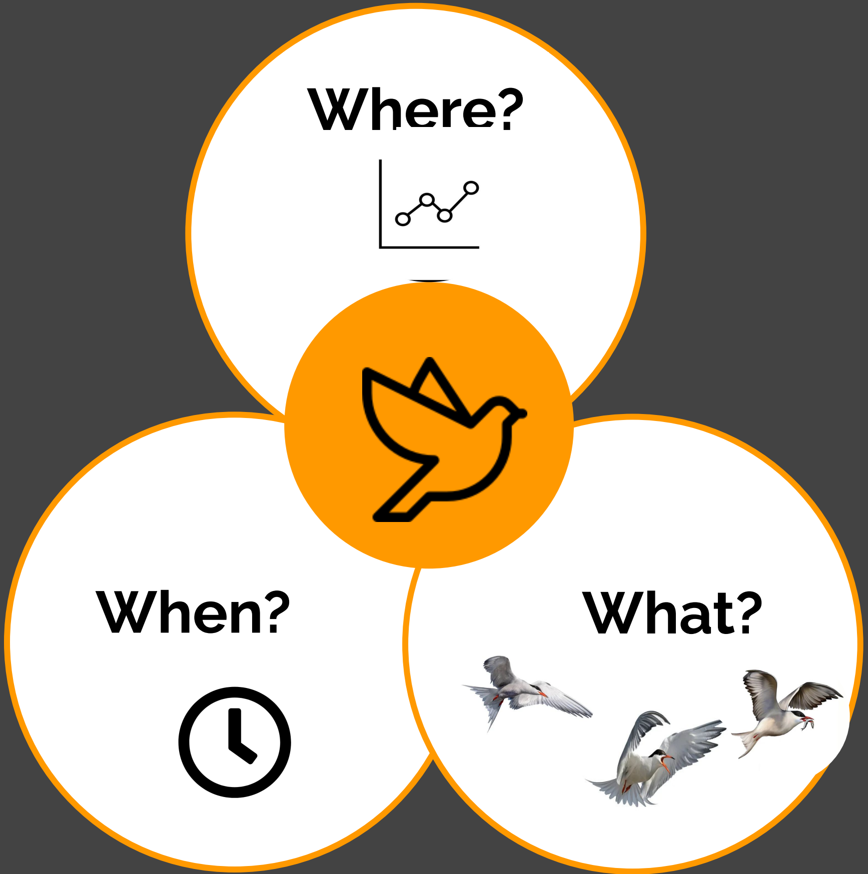

By-catch

# – Background

**The study of animal movement is an active area of research in statistical ecology due to advances in technology**

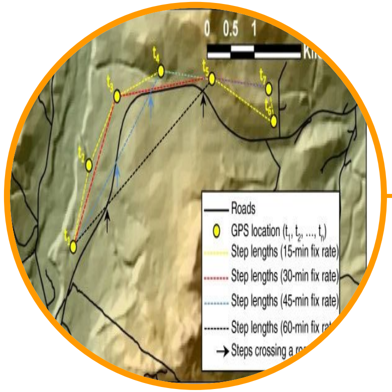




# Background



# — Problem

- Behaviours inferred from movement data are rarely validated in practice




- Loss of behavioural information useful for analysis due to reduction of data points

# What questions did I ask?

- Can we validate inferred behaviours for effective conservation management?
- Can we subsample movement data with minimal loss of information?

Chapter 4

## Assessing visual tracking data

Chapter 5

## Validating inferred behaviours from tracking data

Akeresola et al., "Validating hidden Markov models for seabird behavioural inference", *Ecology and Evolution*, 2024.

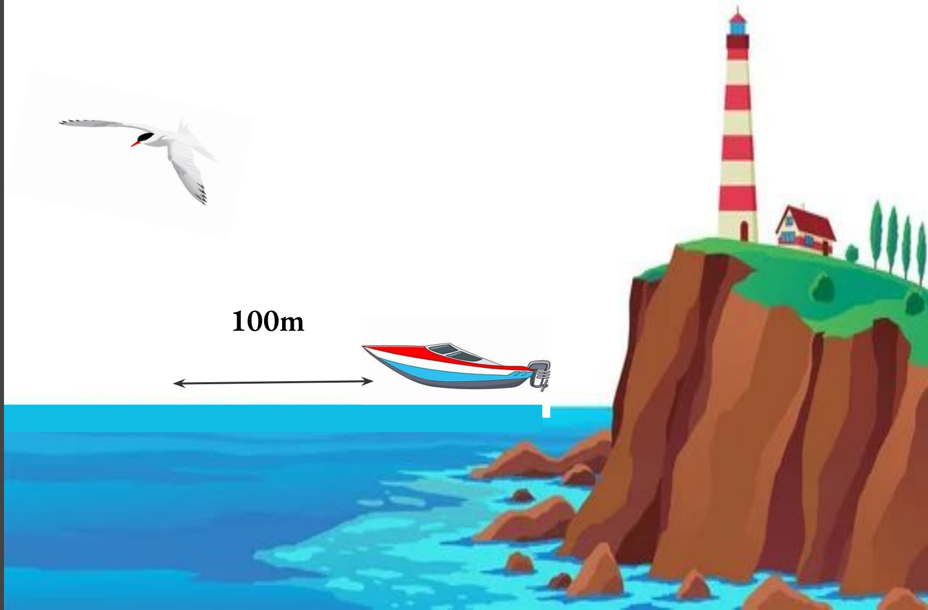
# — Visual Tracking Data

- Sampling interval 1-second
- observed behavioural data

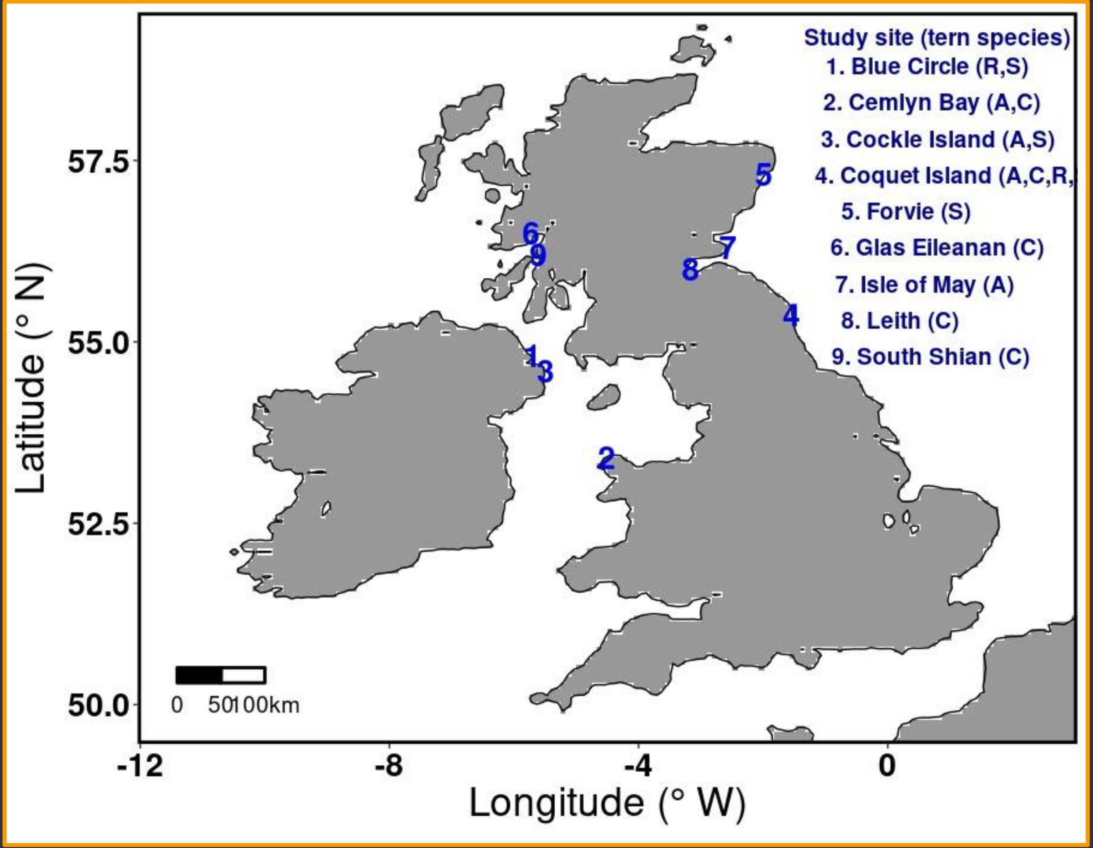
+

- boat GPS location data

Perrow et al., "Visual tracking from a rigid-hulled inflatable boat to determine foraging movements of breeding terns.", *Journal of Field Ornithology*, 2011



# Study sites and species




Arctic, A

Common, C

Roseate, R

Sandwich, S

# Workflow

STEP 1

Overview of **hidden Markov models (HMMs)**

STEP 2

**Assess** boat tracks as proxies for terns tracks

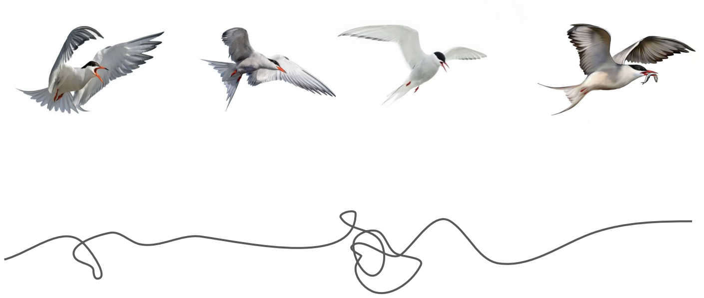
STEP 3

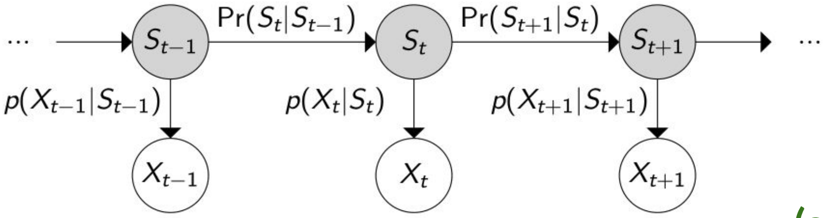
**Validate** HMM-inferred behaviours of terns

# —Hidden Markov model (HMM)

## A time series model made up of two processes

- Markov relationship between hidden sequence of states
- observed and hidden states (Xcurrent|Scurrent)






hidden (state)

observed (state-dependent)

Zucchini, W., MacDonald, I. L. & Langrock, R., "Hidden Markov Models for Time Series: An Introduction Using R", *Chapman and Hall,* 2016.

# —HMMs

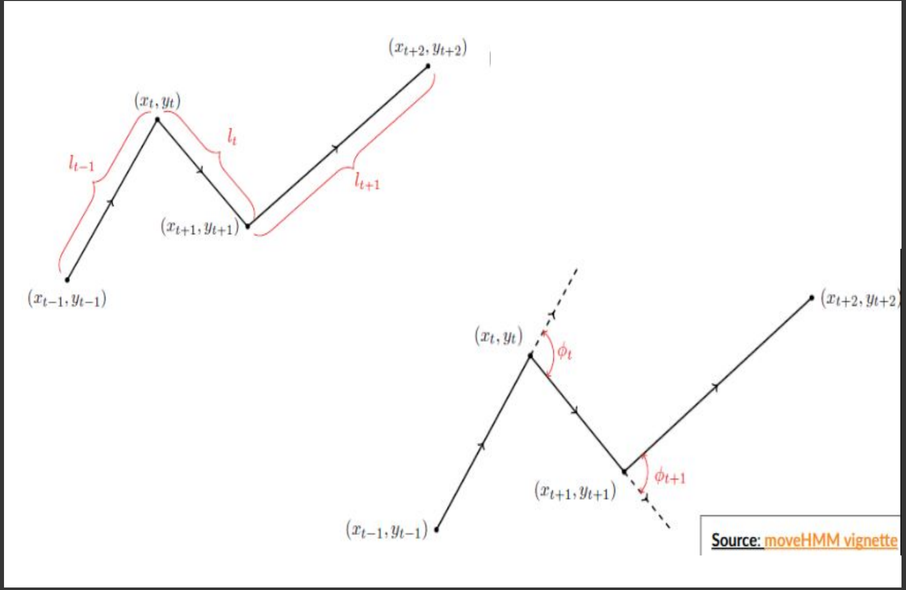
## ~ State process

Specified via the **transition probability matrix**

$$
\begin{array}{ccc}
 & \text{State 1 (forage)} & \text{State 2 (travel)} \\
\begin{array}{c}\text{State 1 (forage)} \\ \text{State 2 (travel)}\end{array} &
\multicolumn{2}{c}{\begin{pmatrix} Pr(S_{11}) & Pr(S_{12}) \\ Pr(S_{21}) & Pr(S_{22}) \end{pmatrix}}
\end{array}
$$

## ~ Observed process

Specified via the **distributions** of step length and turning angles



# Workflow

STEP 1

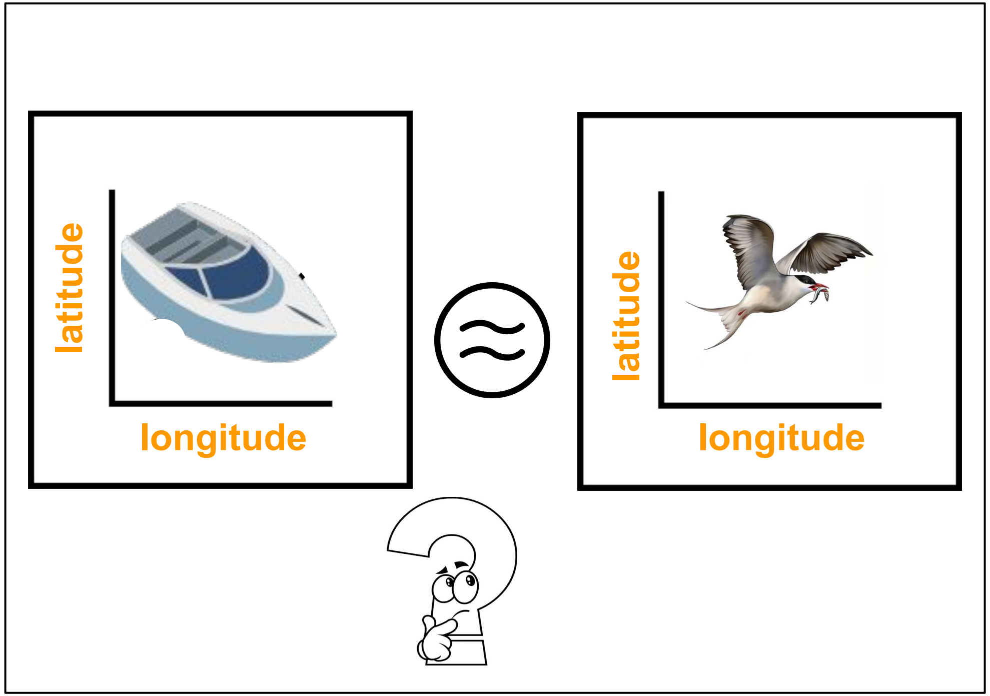
Overview of HMMs

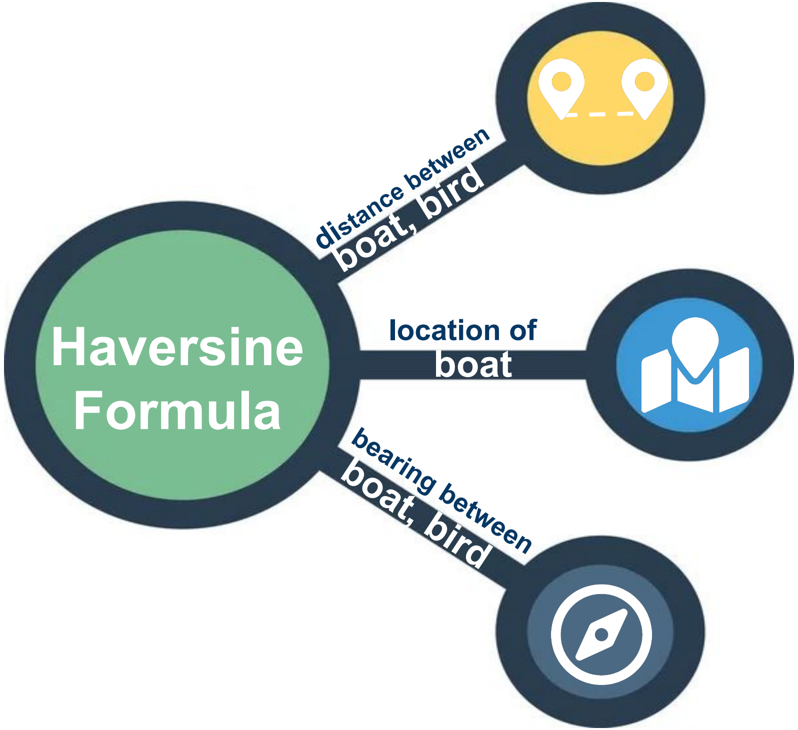
STEP 2

Assess boat tracks as proxies for terns tracks

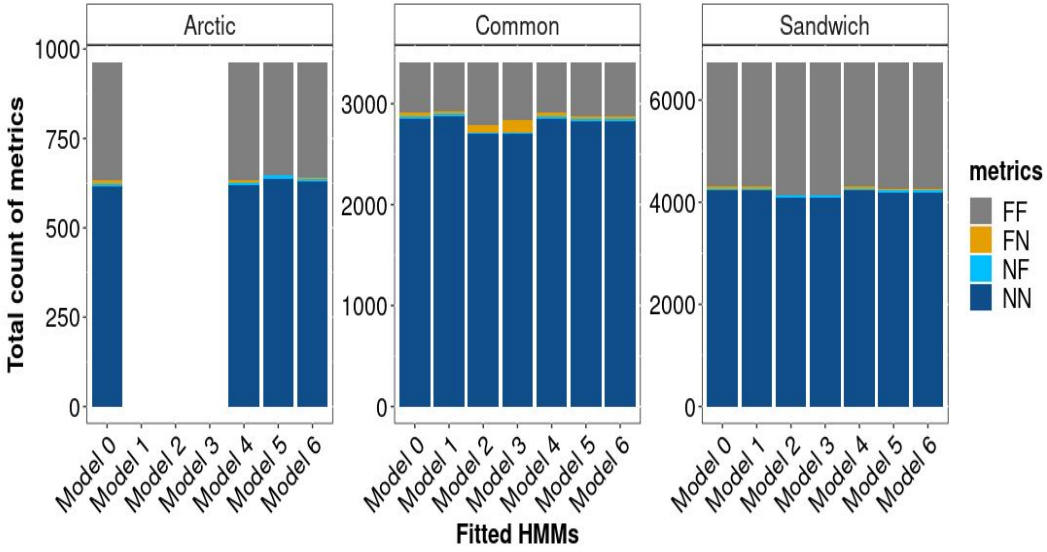
# — Assessing boat location data

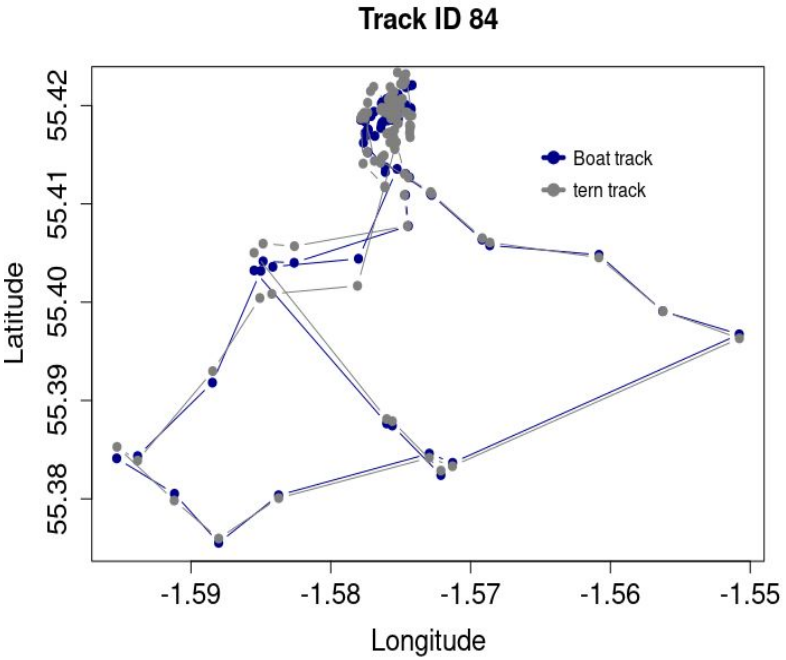
3 Arctic, 3 Common, 1 Roseate and 6 Sandwich terns
(Coquet colony, 2009)



# — Assessing boat location data




Confusion matrix metrics of behaviours inferred from 2-state(foraging (F), not-foraging (N)) HMMs fitted to boat location and approximate location data of terns in Coquet colony during chick-rearing, 2009. FF-true positive, FN -false negative, NF-false positive, NN - true negative

Comparing boat track and approximate tern track from Coquet colony

# Workflow

**STEP 1**

**Overview of hidden Markov models (HMMs)**

**STEP 2**

**Assess boat tracks as proxies for terns tracks**

**STEP 3**

**Validate HMM-inferred behaviours of terns**

# Validation results

- Overall performance of HMMs in decoding behavioural states (f1-score)
- Correctly decoded foraging behavioural state during chick-rearing

Arctic tern
64-74%
58-70%
3 colonies

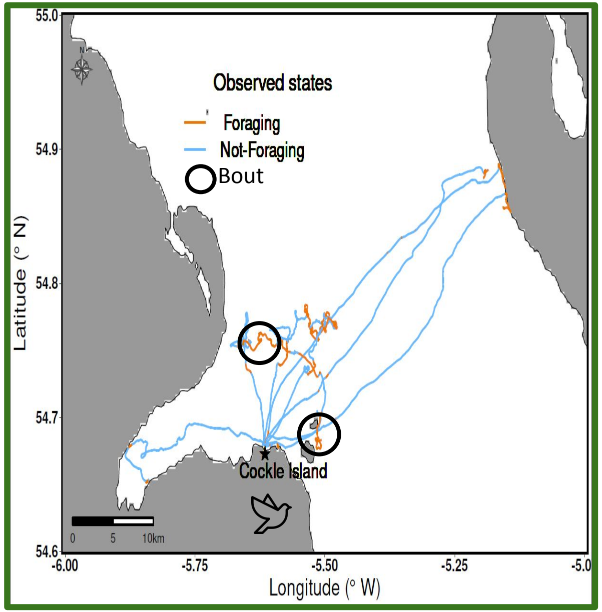
Common tern
76-84%
70-90%
5 colonies

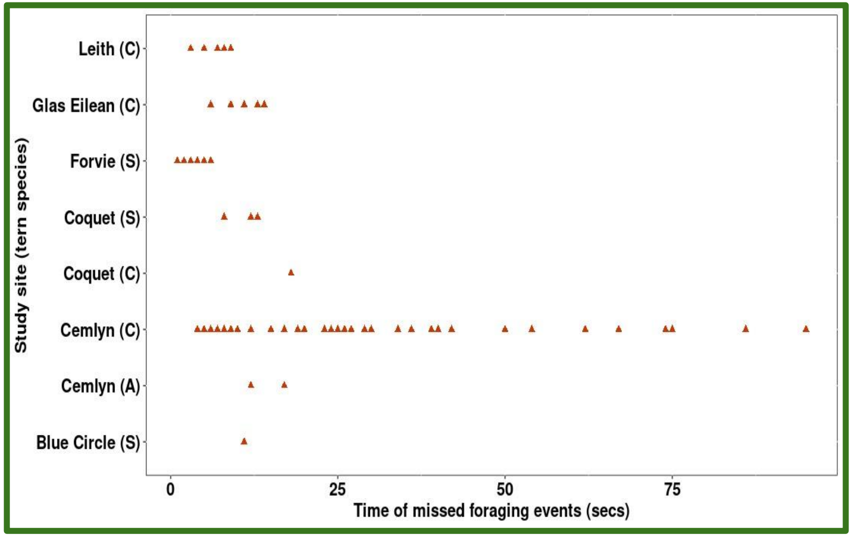
Roseate tern
76%
86%
1 colony

Sandwich tern
72-88%
74-91%
4 colonies

# — Foraging event

A bout where foraging behavioural activity occur within the tracking data




Observed foraging events completely missed from decoded behavioural states across study sites. A, C, S = Arctic, common and Sandwich tern

# What does our result tells us?

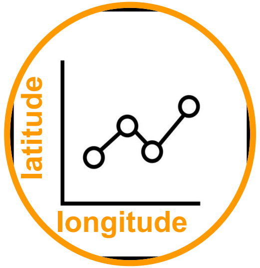
- Visual tracking data can be used as proxies for actual GPS data of birds
- Visual tracking method can be used for identifying important conservation areas
- Increased confidence in unvalidated inferred behaviours from telemetry data
- Conservation-relevant behaviours inferred using HMMs can inform the planning and design of designated protected areas

Chapter 6

# Subsampling animal movement data

## Nyquist-Shannon sampling theorem (NSST) [1, 2]

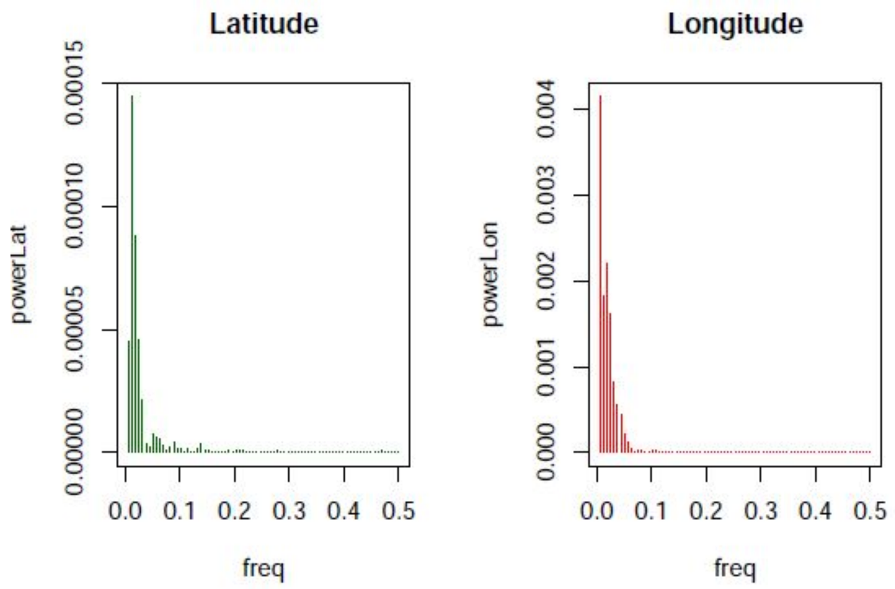
- Describes how to sample  a continuous-time signal (or subsample a discrete-time signal) in a way that there is little or no information loss
  - Useful for selecting frequencies to subsample movement trajectories

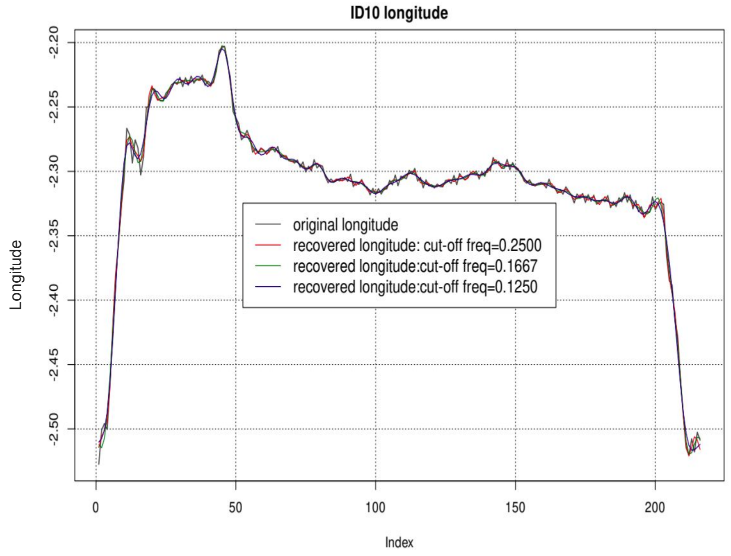



[1] Shannon, Claude E. "Communication in the presence of noise." *Proceedings of the IRE* 37.1 (2006): 10-21.
[2] Nathan et al. “Big-data approaches lead to an increased understanding of the ecology of animal movement.” *Science*. 2022.

# NSST workflow

- Represent movement tracks in frequency domain
- Discard high frequencies at a cut-off frequency of B'< 0.5. (low-pass filter)
- Then subsample, working at a frequency fs'= B'<0.5
- Recover filtered movement tracks in time domain



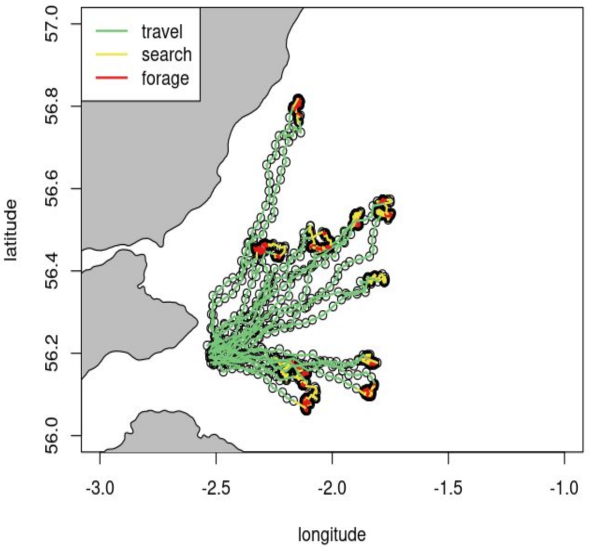



# — Goal

- Understand movement data from the frequency domain perspective
- Investigate how NSST approach minimises information loss compared to thinning approach
- Assess how the choice of cut-off frequency, B', in NSST impacts behavioural inference

# – Simulation study: central place foragers

~simulated tracks with 3 behavioural states at sampling interval of 5-mins~

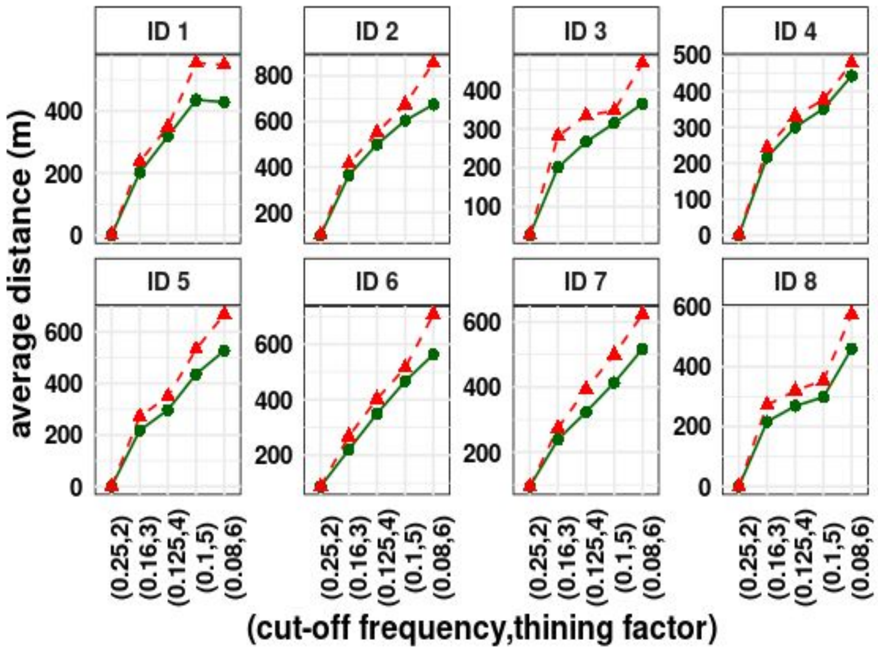



- **Thinning approach:**
  factors: 2, 3, 4, 5 and 6

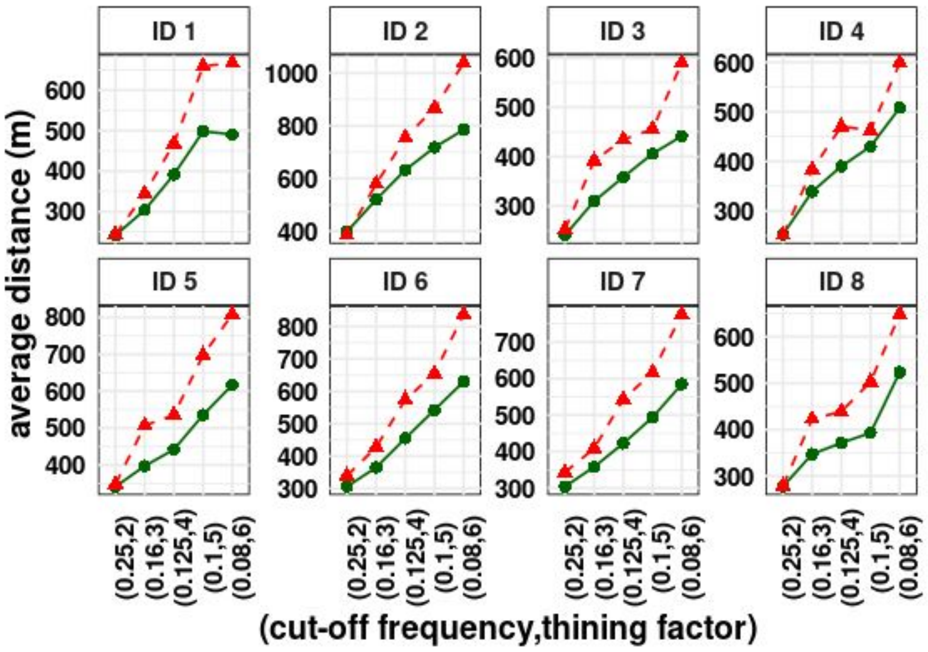
- **NSST approach:**
  cut-off frequencies: 0.25, 0.16, 0.125, and 0.08

# — Result: Comparing thinning and NSST approach

~Scenario 1: no high frequency component (B' = 0.25)




~Scenario 2: some high frequency component



# — Result: comparing thinning (TI) to NSST (FTI) approach

Overall classification accuracy (%) of 3-state complete pooling HMM fitted to simulated tracks.

| | Scenario 1 | | Scenario 2 | |
|---|---|---|---|---|
| **(cf, tf)** | **FTI** | **TI** | **FTI** | **TI** |
| **(0.25,2)** | 88.52 | 88.52 | 88.52 | 89.92 |
| **(0.16,3)** | 89.92 | 88.03 | 89.92 | 82.79 |
| **(0.12,4)** | 86.97 | 88.28 | 86.97 | 86.80 |
| **(0.10,5)** | 84.10 | 82.62 | 84.10 | 82.70 |
| **(0.08,6)** | 82.46 | 74.75 | 82.46 | 82.46 |

*cf = cut-off frequency, tf = thinning factor*

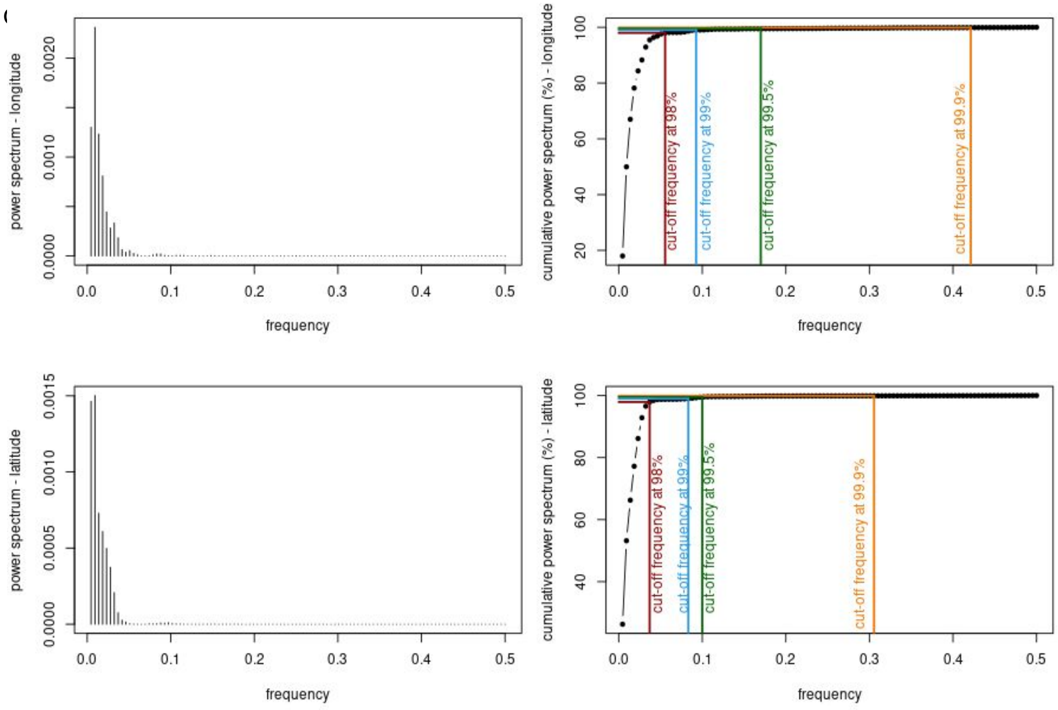
**NSST approach:**

- Behavioural information is preserved across both scenarios
- Higher accuracy in inferring behaviour most times within the two scenarios

**Thinning approach:**

- Decrease in accuracy of behavioural inference from scenarios 1 to 2
- lower accuracy in inferring behaviour most times within the two scenarios

# — Result: choice of cutoff frequency in NSST approach




**Frequency domain representation (left panel) and corresponding cut-off frequency based on high quantiles of of power distribution (right panel) for longitude (top) and latitude (bottom) locations**

Accuracy of 3-state HMM fitted to tracks subsampled at chosen cut-off frequencies

| | Overall accuracy (%) | Travel (%) | Forage (%) | Search (%) |
|---|---|---|---|---|
| Original track | 95.83 | 98.17 | 96.67 | 92.43 |
| Q-99.9% | 94.44 | 98.17 | 92.35 | 81.22 |
| Q-99.5% | 88.89 | 93.48 | 87.27 | 71.12 |
| Q-99% | 80.56 | 93.48 | 83.86 | 73.36 |
| Q-98% | 53.70 | 87.50 | 71.96 | 42.91 |

# – What does our results tells us?

- **NSST provides a better reconstruction of movement data from its subsampled version**
- **We lose more information with the thinning approach.**
- **The use of NSST minimises the loss of behavioural information due to subsampling**

  Animal movement data will have some high-frequency components. As such, using NSST, we can
  - reduce large volumes of movement tracks for computational ease
  - preserve or reduce information loss useful for analysis

- **Quantiles of the frequency distribution can be used as a guide to inform the choice of the cut-off frequency**

  high quantiles of the spectrum distribution are preferable in terms of behavioural inference

# Conclusions

- HMMs are useful for inferring conservation-relevant behaviours
- The frequency domain perspective provides useful insights on how sampling frequency impacts behavioural inference